

Financial Econometrics I

Linear AR and MA Models

(Lecture 3)

Winter Semester 2025/2026 by Lukas Vacha

* If viewed in .pdf format - for full functionality use *Mathematica 7* (or higher) notebook (.nb) version of this .pdf

Stationarity

Linear AR and MA models are cornerstones of time series analysis. They provide a general framework for studying stationary processes. Every covariance (second-order) stationary process is either a linear process or can be transformed to a linear process.

What are the properties of covariance stationary process?

Autoregressive Models

AR (1) model

$$r_t = \phi_0 + \phi_1 r_{t-1} + \epsilon_t, \quad \epsilon_t \sim \text{WN}(0, \sigma^2)$$

Simple autoregressive model of order 1, or simply AR(1) model has the same form as simple linear regression model, where r_t is dependent and r_{t-1} explanatory variable, but has different properties. Mean and variance conditional on past returns are: $E(r_t | r_{t-1}) = \phi_0 + \phi_1 r_{t-1}$, $\text{Var}(r_t | r_{t-1}) = \text{Var}(\epsilon_t) = \sigma_\epsilon^2$.

Thus given past return r_{t-1} , current return is centered around $\phi_0 + \phi_1 r_{t-1}$ with variability σ_ϵ^2 .

AR(1) to MA(∞) - Koyck Transformation

We can rewrite a **stationary** AR(p) model as a MA(∞):

$$\begin{aligned} r_t &= \phi r_{t-1} + \epsilon_t \\ r_t(1 - \phi L) &= \epsilon_t \\ r_t &= \frac{\epsilon_t}{1 - \phi L} \\ r_t &= \epsilon_t + \phi \epsilon_{t-1} + \phi^2 \epsilon_{t-2} + \dots \\ r_t &= \frac{1}{\phi(L)} \epsilon_t = \phi^{-1}(L) \epsilon_t \end{aligned}$$

Properties of AR(1) model

Assuming that series are weakly stationary, we have $E(r_t) = \mu$, $\text{Var}(r_t) = \gamma_0$ and $\text{Cov}(r_t, r_{t-j}) = \gamma_j$, where μ and γ_0 are constants. Thus (unconditional) mean and variance of the AR(1) series is:

$$E(r_t) = \mu = \frac{\phi_0}{1-\phi_1}, \quad \text{Var}(r_t) = \gamma_0 = \frac{\sigma_\epsilon^2}{1-\phi_1^2}.$$

The mean of r_t exist if $\phi_1 \neq 1$, and is zero if and only if $\phi_0 = 0$.

Necessary and sufficient condition for the AR(1) model to be weakly stationary is $|\phi_1| < 1$.

ACF of r_t satisfies: $\rho_\ell = \phi_1 \rho_{\ell-1}$ for $\ell \geq 1$, but as $\rho_0 = 1$, $\rho_\ell = \phi_1^\ell$, and ACF of weakly stationary AR(1) series decays exponentially with rate ϕ_1

How do we compute the $E(r_t)$?

$$\begin{aligned} E(r_t) &= E(\phi_0 + \phi_1 r_{t-1}) \\ &= \phi_0 + \phi_1 E(r_{t-1}) \\ E(r_t) &= \phi_0 + \phi_1(\phi_0 + \phi_1 E(r_{t-2})) \\ &= \phi_0 + \phi_1 \phi_0 + \phi_1^2 E(r_{t-2}) \\ &\vdots \\ E(r_t) &= \phi_0(1 + \phi_1 + \phi_1^2 + \dots) + \phi_1^\infty y_0 \end{aligned}$$

As long as the model is stationary:

$$\phi_1^\infty = 0 \quad \longrightarrow \quad E(r_t) = \phi_0(1 + \phi_1 + \phi_1^2 + \dots) = \frac{\phi_0}{1-\phi_1}$$

How do we compute the $\text{Var}(r_t)$?

Inverting the AR(1) to the MA(∞):

$$r_t = \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_1^2 \epsilon_{t-2} + \dots$$

$$\text{Var}(r_t) = \gamma_0 = E(r_t - E(r_t))(r_t - E(r_t))$$

since $\phi_0 = 0$, $E(r_t) = 0$, and

$$\begin{aligned} \text{Var}(r_t) &= E((r_t)(r_t)) \\ &= E((\epsilon_t + \phi_1 \epsilon_{t-1} + \phi_1^2 \epsilon_{t-2} + \dots)(\epsilon_t + \phi_1 \epsilon_{t-1} + \phi_1^2 \epsilon_{t-2} + \dots)) \\ &= E(\epsilon_t^2 + \phi_1^2 \epsilon_{t-1}^2 + \phi_1^4 \epsilon_{t-2}^2 + \dots + \text{cross-products}) \\ &= \sigma_\epsilon^2 + \phi_1^2 \sigma_\epsilon^2 + \phi_1^4 \sigma_\epsilon^2 + \dots \\ &= \sigma_\epsilon^2(1 + \phi_1^2 + \phi_1^4 + \dots) \\ &= \frac{\sigma_\epsilon^2}{1-\phi_1^2} \end{aligned}$$

How do we compute the $\text{Cov}(r_t, r_{t-s})$?

$$\begin{aligned} \gamma_1 &= \text{Cov}(r_t, r_{t-1}) = E(r_t - E(r_t))(r_{t-1} - E(r_{t-1})) \\ \gamma_1 &= E((\epsilon_t + \phi_1 \epsilon_{t-1} + \phi_1^2 \epsilon_{t-2} + \dots)(\epsilon_{t-1} + \phi_1 \epsilon_{t-2} + \phi_1^2 \epsilon_{t-3} + \dots)) \\ &\vdots \\ \gamma_1 &= \phi_1 \sigma_\epsilon^2 + \phi_1^3 \sigma_\epsilon^2 + \phi_1^5 \sigma_\epsilon^2 + \dots \\ &= \frac{\phi_1 \sigma_\epsilon^2}{1-\phi_1^2} \end{aligned}$$

Second autocovariance, $\gamma_2 = \frac{\phi_1^2 \sigma^2}{1 - \phi_1^2}$, and autocovariances of higher orders, $\gamma_s = \frac{\phi_1^s \sigma^2}{1 - \phi_1^2}$, are computed similarly.

How do we compute the ACF function ?

$$\begin{aligned}\rho_0 &= \frac{\gamma_0}{\gamma_0} = 1 \\ \rho_1 &= \frac{\gamma_1}{\gamma_0} = \frac{\frac{\phi_1 \sigma^2}{1 - \phi_1^2}}{\frac{\sigma^2}{1 - \phi_1^2}} = \phi_1 \\ &\vdots \quad \quad \quad \vdots \\ \rho_s &= \frac{\gamma_s}{\gamma_0} = \frac{\frac{\phi_1^s \sigma^2}{1 - \phi_1^2}}{\frac{\sigma^2}{1 - \phi_1^2}} = \phi_1^s\end{aligned}$$

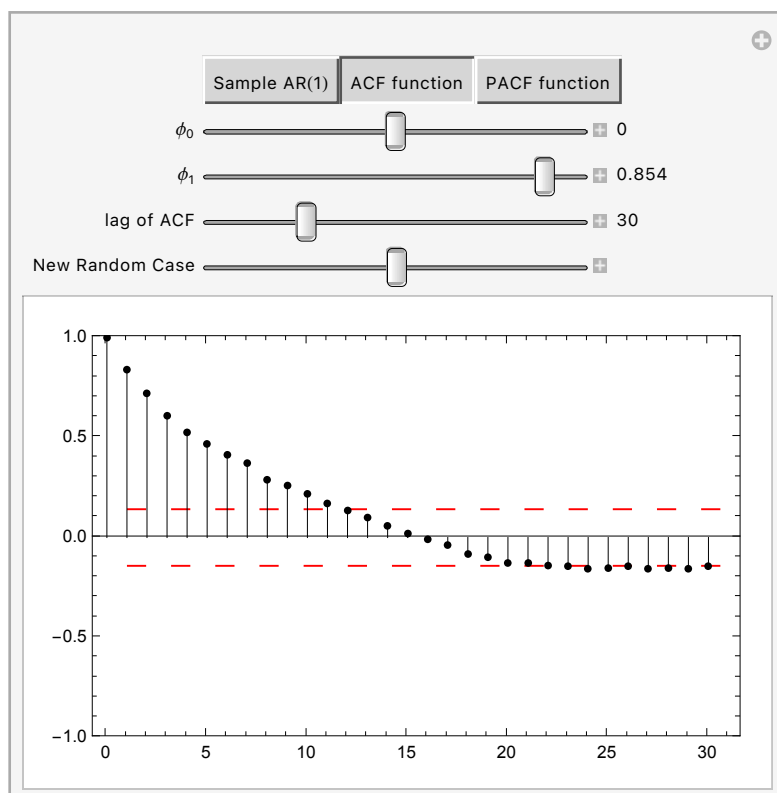
Partial correlation function (PACF)

The j 'th partial autocorrelation, ϕ_{jj} , is related to the coefficient on r_{t-j} of a regression of r_t on $r_{t-1}, r_{t-2}, \dots, r_{t-j}$. Thus for an AR(p), the $j + 1$ th and higher partial autocorrelations are zero. The PACF was defined by Box and Jenkins. It serves as an identification tool for ARMA processes.

The partial autocorrelations may be estimated by fitting successively autoregressive models of orders 1, 2, 3, ... by least squares and picking out the estimates $\hat{\phi}_{11}, \hat{\phi}_{22}, \hat{\phi}_{33}, \dots$ of the last coefficient fitted at each stage.

Examples of AR(1) artificial processes and their ACFs and PACFs

i.e. set $\phi_1 = -0.99$



AR(2) model

Simple autoregressive model of order 2, or simply AR(2) model is represented by :

$$r_t = \phi_0 + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \epsilon_t,$$

where $\{\epsilon_t\}$ is assumed to be a white noise $\epsilon_t \sim \text{WN}(0, \sigma^2)$.

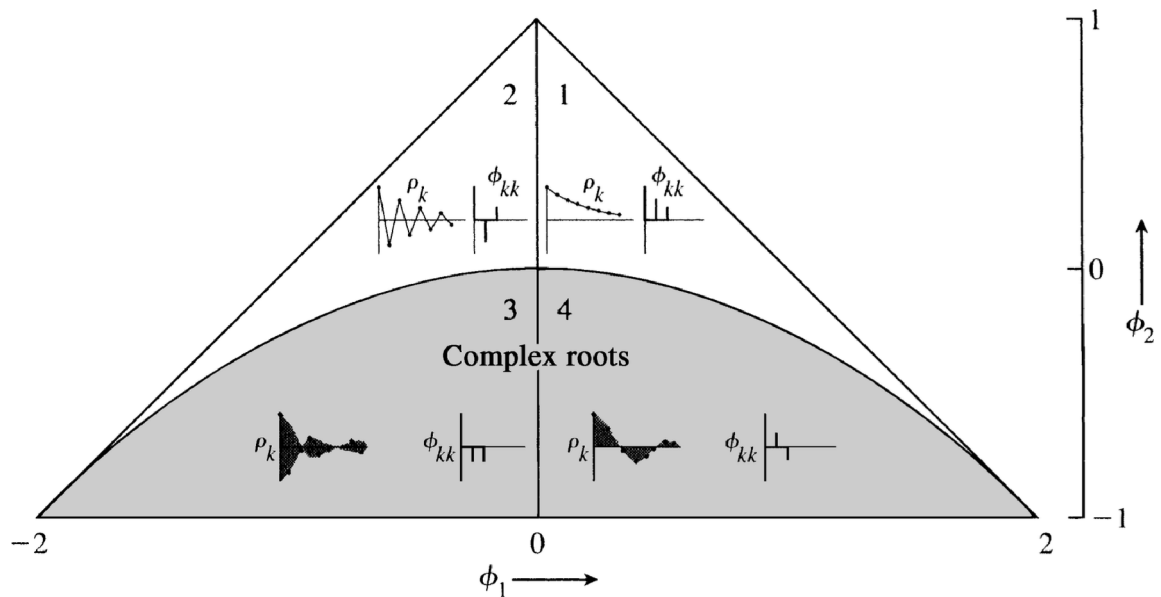
Properties of AR(2) model

Mean of the AR(2) model is: $E(r_t) = \frac{\phi_0}{1-\phi_1-\phi_2}$, provided $\phi_1 + \phi_2 \neq 1$

For a stationary AR(2) process the characteristic roots of the second order difference equation $(1 - \phi_1 L - \phi_2 L^2) = 0$ **lie outside the unit circle**.

Solution to second order polynomial equation $x^2 - \phi_1 x - \phi_2 = 0$ are characteristic roots of AR(2) model. For stationary AR(2) the characteristic roots **lie inside the unit circle**.

If they are real numbers, AR(2) model is just AR(1) model operating on top of another AR(1). In this case, ACF is mixture of two exponential decays. If the characteristic roots are complex, ACF is damping sine and cosine waves. Complex characteristic roots are important in business cycles.



Autocorrelation and partial autocorrelation functions for various stationary AR(2) models (Source: Stralkowski, 1968).

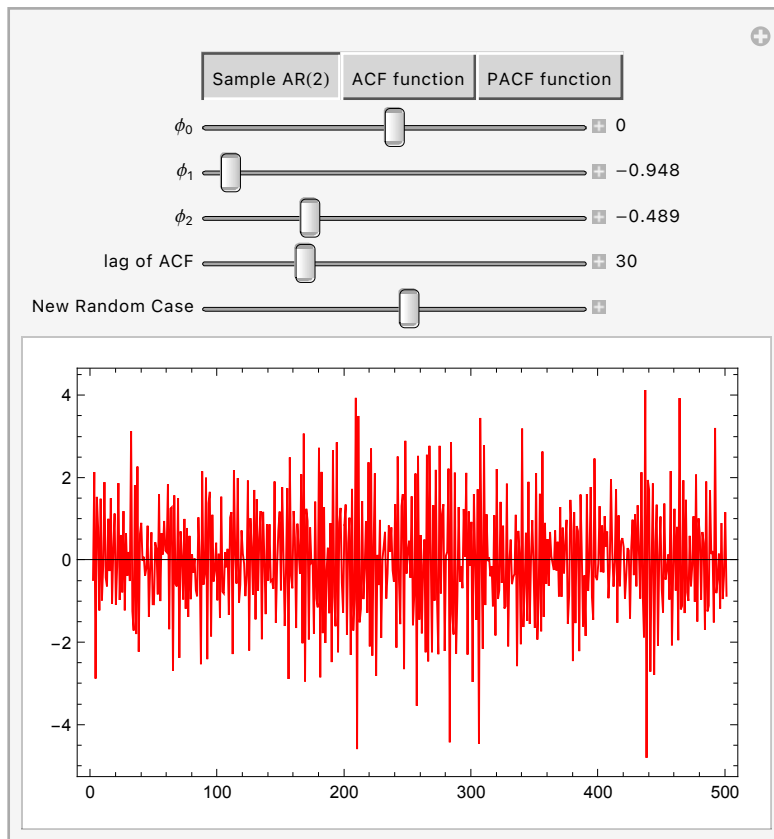
Examples of AR (2) artificial processes and their ACFs

Set $\phi_1 = 0.6$ and $\phi_2 = -0.4$ for example of random stationary AR(2) process with complex characteristic roots.

Second order difference equation is $(1 - 0.6L + 0.4L^2)r_t = \epsilon_t$, and ACF is damping sine and cosine waves

Set $\phi_1 = -0.2$ and $\phi_2 = 0.35$ for example of random stationary AR(2) process with real characteristic roots. ACF decays exponentially.

Note that $\phi_1 + \phi_2$ must be less than 1, otherwise AR(2) would not be stationary



AR(p) model

Simple autoregressive model of order p , or simply AR(p) model is represented by:

$$r_t = \phi_0 + \sum_{i=1}^p \phi_i r_{t-i} + \epsilon_t,$$

where p is non-negative integer and $\epsilon_t \sim \text{WN}(0, \sigma^2)$, is generalization of AR(1) to AR(p) model.

Or equivalently (without intercept):

$$\phi(L) r_t = \epsilon_t,$$

where $\phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p$.

(L denotes a lag operator such that $L X_t = X_{t-1}$.)

Properties of AR (p) model

Mean of the AR(p) model is: $E(r_t) = \frac{\phi_0}{1 - \phi_1 - \dots - \phi_p}$, provided that denominator is zero.

Characteristic equation of the model is $x^p - \phi_1 x^{p-1} - \dots - \phi_p = 0$, and if all roots of the equation are **less than one in modulus**, series is stationary.

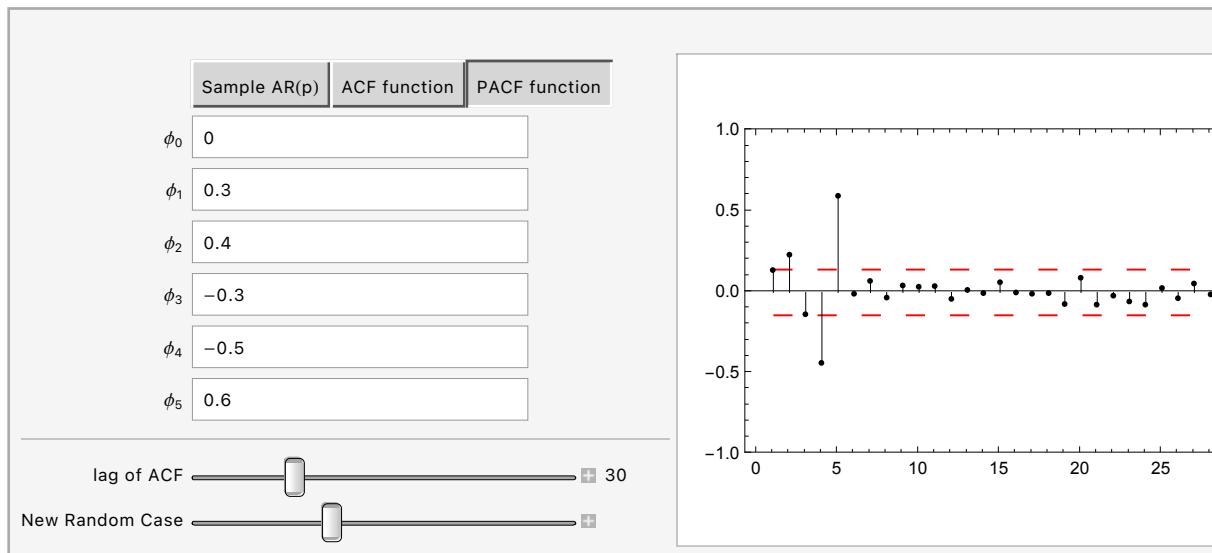
PACF of AR(p) model

Estimate of the second lag of PACF shows added contribution of r_{t-2} to r_t over the AR(1) model $r_t = \phi_1 r_{t-1} + e_t$, the lag-3 PACF shows the added contribution of r_{t-3} to AR(2), and so

on. Thus for AR(p) model, lag- p should not be zero, but all other lag- j such as $j > p$ should be close to zero. Thus we can identify AR(p) process with PACF function.

Examples of AR (p) artificial processes and their ACFs

Note that $\phi_1 + \phi_2 + \dots + \phi_p$ must be less than 1, otherwise AR(p) would not be stationary



Moving average (MA) models

MA(1) model

Simple Moving average model of order 1, or MA(1) model is represented by:

$$r_t = \theta_0 + \epsilon_t + \theta_1 \epsilon_{t-1},$$

where θ_0 is constant and $\{\epsilon_t\}$ is a white noise series.

Properties of MA(1) model

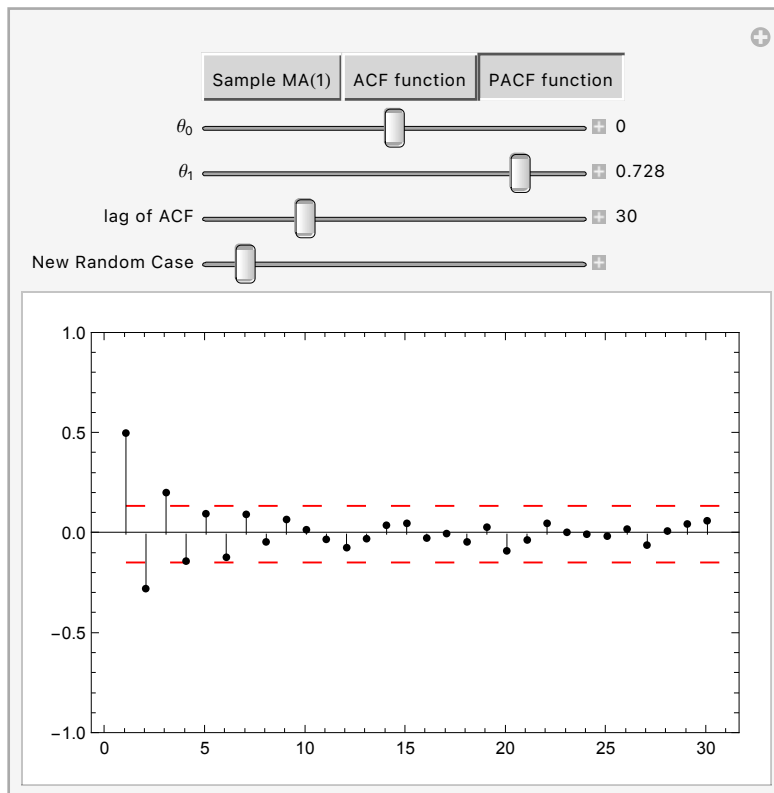
MA models are always weakly stationary because they are finite linear combination of a white noise sequence.

$$E(r_t) = \theta_0, \text{Var}(r_t) = (1 + \theta_1^2) \sigma^2,$$

the lag-1 ACF is not zero. For $\rho_1 = \frac{\theta_1}{1 + \theta_1^2}$ and $\rho_k = 0$ for $k > 1$.

Example of MA(1) artificial processes

Out[]=



Invertibility condition MA(1) to AR(∞)

Consider zero mean MA(1) in the form

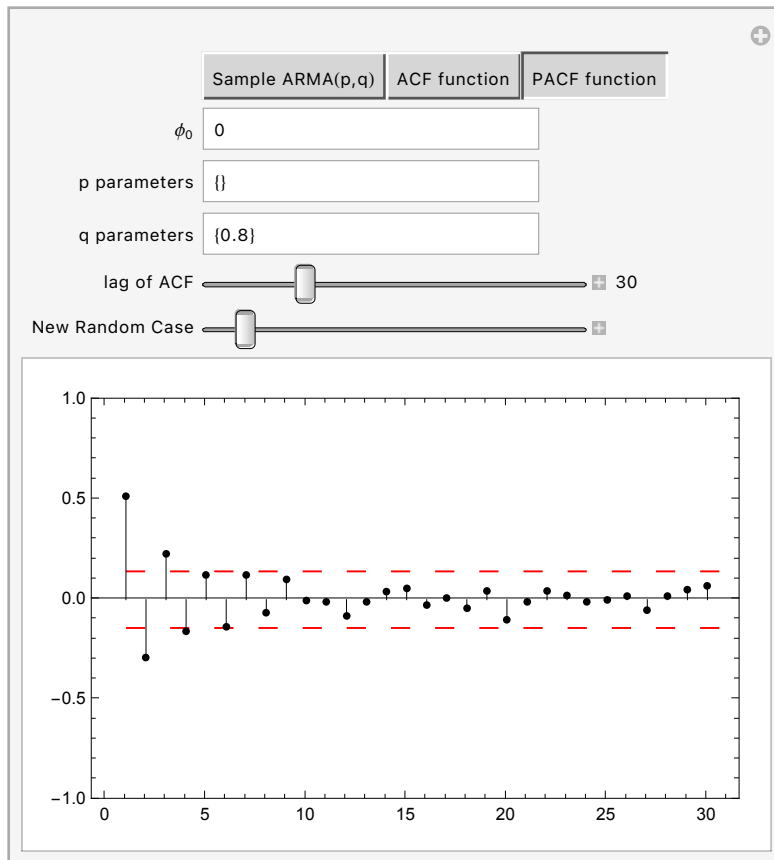
$$r_t = \epsilon_t + \theta \epsilon_{t-1}$$

$$\frac{r_t}{1 + \theta L} = \epsilon_t$$

$$r_t - \theta r_{t-1} + \theta^2 r_{t-2} \dots = \epsilon_t$$

$$r_t = \epsilon_t + \theta r_{t-1} - \theta^2 r_{t-2} + \dots$$

Example MA(1) to AR(∞)



MA(2) model

Simple Moving average model of order 2, or MA(2) model is represented by:

$$r_t = \theta_0 + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2},$$

where θ_0 is constant and $\{\epsilon_t\}$ is a white noise series.

Properties of MA(2) model

MA models are always weakly stationary because they are finite linear combination of a white noise sequence.

$$E(r_t) = \theta_0, \text{Var}(r_t) = (1 + \theta_1^2 + \theta_2^2) \sigma^2,$$

the ACFs of lag 1 and 2 are non-zero, all others are zero.

How do we compute the $\text{Var}(r_t)$

In case $\theta_0 = 0$, i.e. $E(r_t) = 0$,

$$\begin{aligned} \text{Var}(r_t) &= E((r_t)(r_t)) \\ &= E((\epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2})(\epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2})) \end{aligned}$$

since $\text{Cov}(\epsilon_t, \epsilon_{t-s}) = 0$ for $s \neq 0$, $E(\text{cross-products})=0$, and

$$\begin{aligned}\text{Var}(r_t) &= \gamma_0 = \sigma^2 + \theta_1^2 \sigma^2 + \theta_2^2 \sigma^2 \\ &= (1 + \theta_1^2 + \theta_2^2) \sigma^2\end{aligned}$$

How do we compute the $\text{Cov}(r_t, r_{t-s})$

$$\begin{aligned}\gamma_1 &= \text{Cov}(r_t, r_{t-1}) = E(r_t - E(r_t))(r_{t-1} - E(r_{t-1})) \\ \gamma_1 &= (\theta_1 + \theta_1 \theta_2) \sigma^2\end{aligned}$$

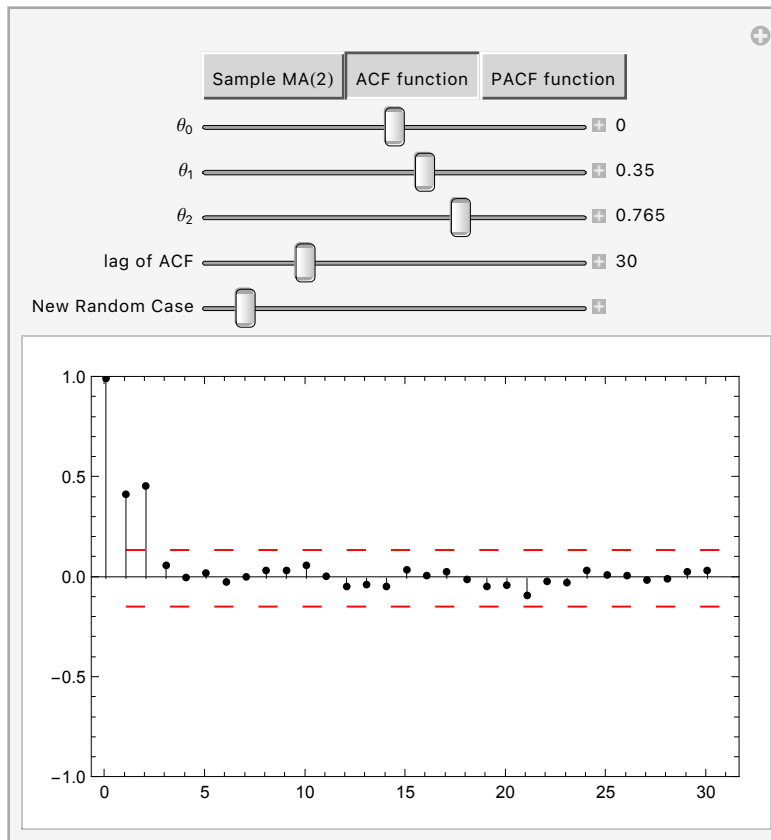
$$\begin{aligned}\gamma_2 &= \theta_2 \sigma^2 \\ \gamma_3 &= 0\end{aligned}$$

autocovariances of higher orders, $\gamma_s = 0$, for $s > 2$.

How do we compute the ACF function:

$$\begin{aligned}\rho_0 &= \frac{\gamma_0}{\gamma_0} = 1 \\ \rho_1 &= \frac{\gamma_1}{\gamma_0} = \frac{(\theta_1 + \theta_1 \theta_2) \sigma^2}{(1 + \theta_1^2 + \theta_2^2) \sigma^2} = \frac{\theta_1 + \theta_1 \theta_2}{1 + \theta_1^2 + \theta_2^2} \\ \rho_2 &= \frac{\gamma_2}{\gamma_0} = \frac{\theta_2 \sigma^2}{(1 + \theta_1^2 + \theta_2^2) \sigma^2} = \frac{\theta_2}{1 + \theta_1^2 + \theta_2^2} \\ &\vdots \quad \quad \quad \vdots \\ \rho_s &= \frac{\gamma_s}{\gamma_0} = 0 \quad \forall s > 2\end{aligned}$$

Examples of MA(2) artificial processes



MA(q) model

Simple Moving average model of order q , or MA(q) model is represented by:

$$r_t = \theta_0 + \epsilon_t + \sum_{i=1}^q \theta_i \epsilon_{t-i},$$

or,

$$r_t - \theta_0 = (1 + \theta_1 L + \theta_2 L^2 + \dots + \theta_q L^q) \epsilon_t = \theta(L) \epsilon_t.$$

where θ_0 is constant and $\{\epsilon_t\}$ is a white noise series, $q > 0$.

Hence, the moving average process can be thought of as the output r_t from a linear filter with transfer function $\theta(L)$ when the input is white noise ϵ_t .

Properties of MA(q) model

MA models are always weakly stationary because they are finite linear combination of a white noise sequence.

$$E(r_t) = \theta_0, \text{Var}(r_t) = (1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2) \sigma^2$$

$$\gamma_s = \begin{cases} (\theta_s + \theta_{s+1} \theta_1 + \theta_{s+2} \theta_2 + \dots + \theta_q \theta_{q-s}) \sigma^2 & s = 1, 2, \dots, q \\ 0 & s > q. \end{cases}$$

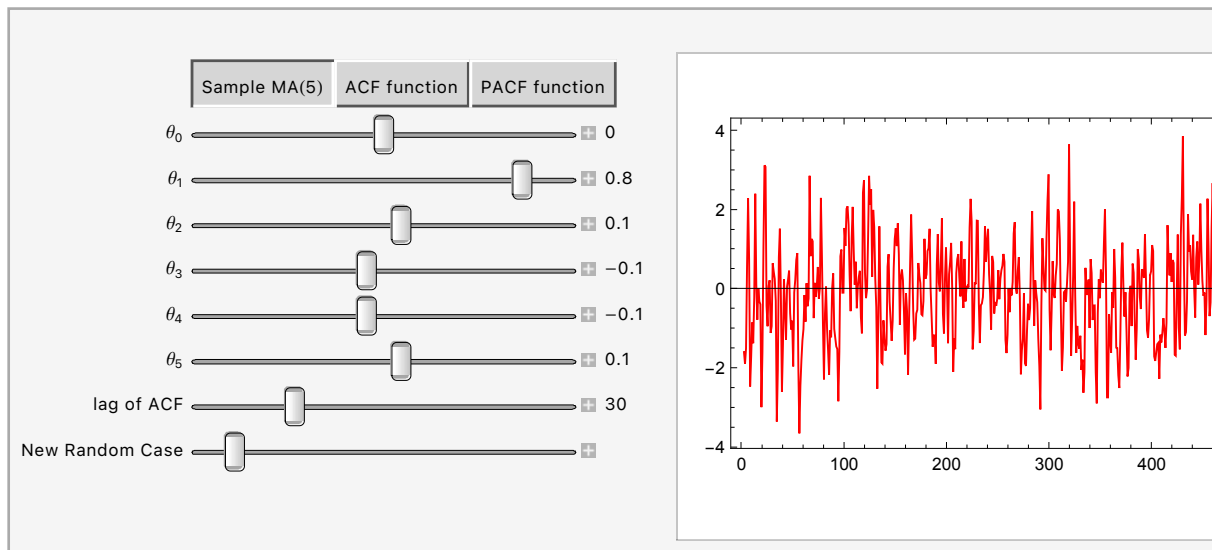
the lag- q ACF is not zero, but $\rho_l = 0$ for all $l > q$.

The Invertibility Condition

Invertibility condition for the MA(q) : The roots of $(1 + \theta_1 L + \dots + \theta_q L^q) = 0$ lie outside the unit circle. **The invertibility condition is mathematically same as the stationarity condition for AR(p) models.**

Examples of MA(q) artificial processes

Example of random MA(5) process - note changes in ACF



Autoregressive Moving-Average (ARMA) models

ARMA models basically combines AR and MA models into a compact form. In returns series, the ARMA models do not seem to be a use full tool, but in volatility modeling, they are considered as highly relevant.

ARMA(1,1)

A time series r_t follows ARMA(1,1) model, if it satisfies:

$$r_t - \phi_1 r_{t-1} = \phi_0 + \epsilon_t + \theta_1 \epsilon_{t-1},$$

where $\{\epsilon_t\}$ is a white noise series, left-hand side is AR(1) model and right-hand side is MA(1) model.

Properties of ARMA(1,1) model

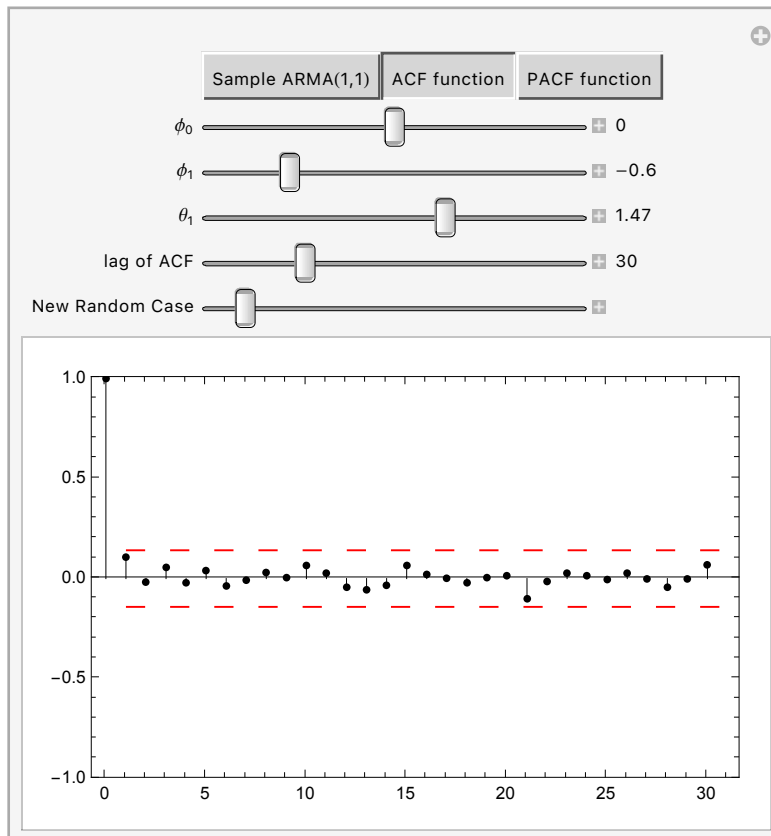
Provided series are weakly stationary, mean is the same as the mean of AR(1), because $E(\epsilon_t) = 0$

$$E(r_t) = \mu = \frac{\phi_0}{1-\phi_1}, \quad \text{Var}(r_t) = \frac{(1-2\phi_1\theta_1+\theta_1^2)\sigma^2}{1-\phi_1^2},$$

and $\phi_1^2 < 1$ condition is needed for stationarity condition.

ACF for stationary ARMA(1,1) series is $\rho_1 = \phi_1 + \frac{\theta_1\sigma^2}{\gamma_0}$, $\rho_\ell = \phi_1\rho_{\ell-1}$ for $\ell > 1$. Thus ACF of ARMA(1,1) behaves very much like the ACF of AR(1), PACF of an ARMA(1,1) does not cut off at any lag.

Examples of ARMA(1,1) artificial processes



General ARMA(p,q) models

A general ARMA(p,q) model is in the form:

$$r_t = \phi_0 + \sum_{i=1}^p \phi_i r_{t-i} + \epsilon_t + \sum_{i=1}^q \theta_i \epsilon_{t-i},$$

where $\{\epsilon_t\}$ is a white noise series and p and q are non-negative integers. The AR and MA models are special cases of the ARMA(p,q) model. Using the back-shift operator, the model is written as:

$$(1 - \phi_1 L - \dots - \phi_p L^p) r_t = \phi_0 + (1 + \theta_1 L + \dots + \theta_q L^q) \epsilon_t$$

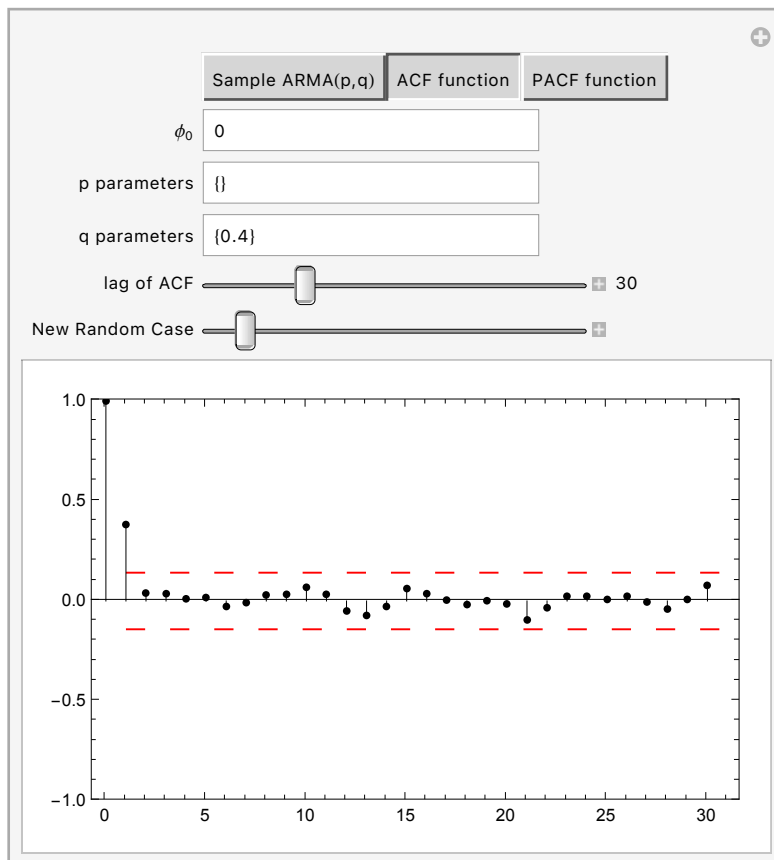
or using the polynomial operators:

$$\phi(L) r_t = \phi_0 + \theta(L) \epsilon_t$$

$$r_t - \phi_0 = \frac{\theta(L)}{\phi(L)} \epsilon_t = \frac{1 + \theta_1 L + \dots + \theta_q L^q}{1 - \phi_1 L - \dots - \phi_p L^p} \epsilon_t.$$

The ARMA process can be thought of as the output r_t from a linear filter, whose transfer function is the ratio of two polynomial operators $\phi(L)$ and $\theta(L)$, when the input is white noise ϵ_t .

Examples of ARMA(p,q) artificial processes



Wold's Decomposition

Wold's decomposition theorem (1938) states that any zero-mean stationary series can be decomposed into the sum of two unrelated processes, a purely deterministic part and a purely stochastic part - an MA (∞):

$$r_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j} + \mu_t,$$

where $\sum_{j=0}^{\infty} \psi_j^2 < \infty$ with $\psi_0 = 1$.

μ_t is called the linearly deterministic component of r_t , it is uncorrelated with ϵ_{t-j} for any j . (it is a linear combination of past values of μ_t with constant coefficients).

The decomposition relies on stable second moments r_t . With finite number of observations

we need to assume that $\psi(L)$ can be expressed as the ratio of two finite-order polynomials:

$$\sum_{j=0}^{\infty} \psi_j L^j = \frac{\theta(L)}{\phi(L)} = \frac{1 + \theta_1 L + \dots + \theta_q L^q}{1 - \phi_1 L - \dots - \phi_p L^p}.$$

Impulse Response function (IRF)

An IRF is the change in the current and expected future values of a random variable conditional on the realization of uncertainty at some point in time. We can define it as:

$$\text{IRF}_j = E_t r_{t+j} - E_{t-1} r_{t+j} \quad \text{for all } j \geq 0$$

If ARMA is defined as $r_t = b(L) \epsilon_t$, then IRF is just $b(L) \epsilon_t$.

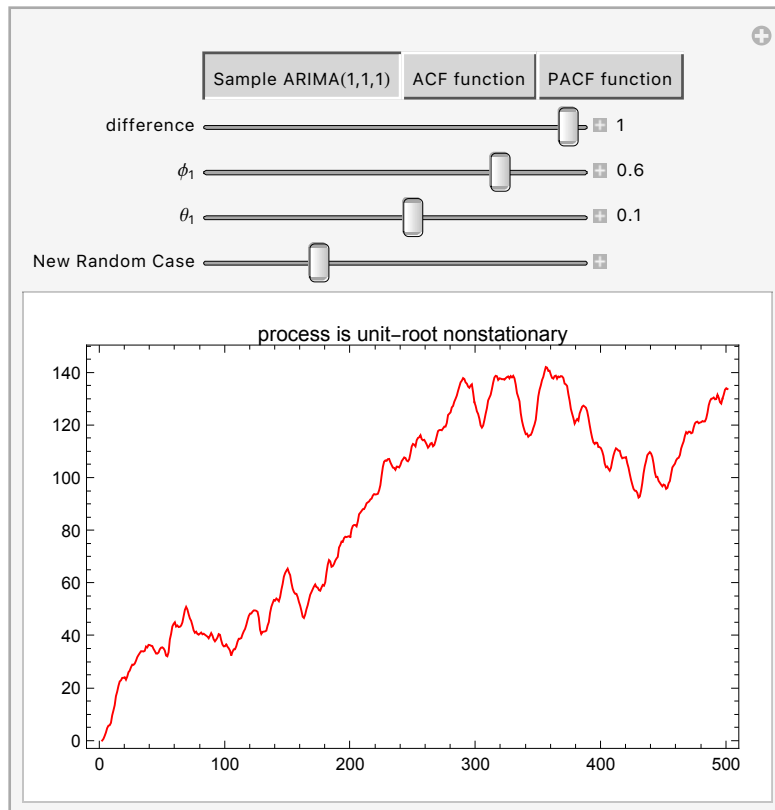
ARIMA (p, d, q) model

If we extend the ARMA model by allowing the AR polynomial to have 1 as a characteristic root, then the model becomes AutoRegressive Integrated Moving-Average (ARIMA) model. ARIMA model is unit-root nonstationary because its AR polynomial has a unit root. Like a random-walk model, an ARIMA model has strong memory. If we want to handle nonstationarity, we use *differencing* as a common approach

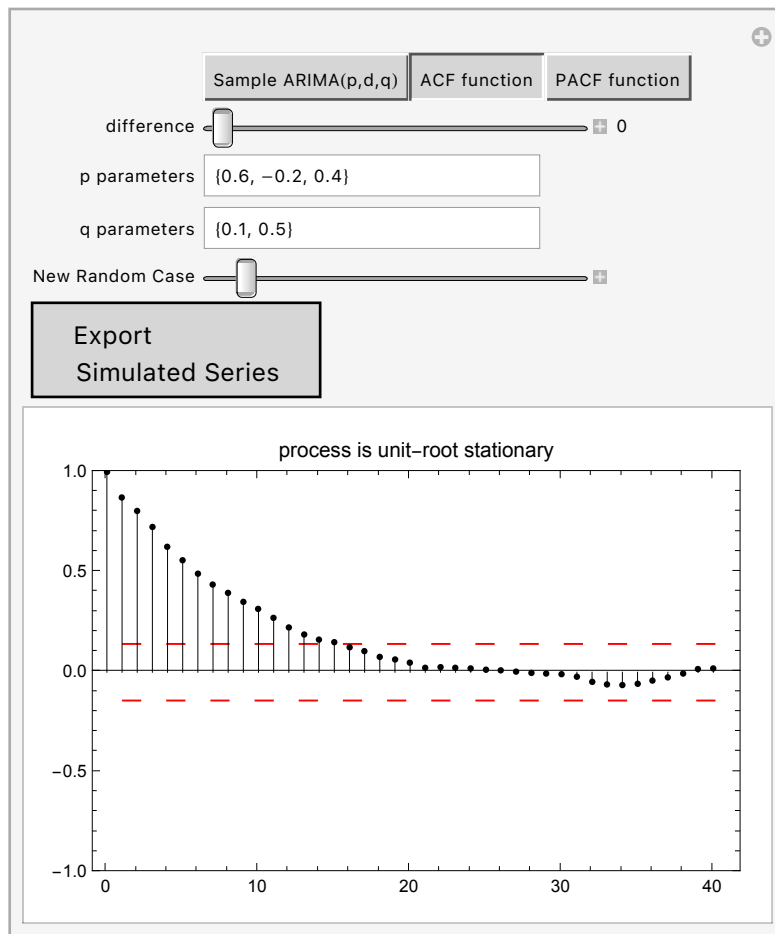
Differencing

Series are said to be ARIMA($p, 1, q$), when the process $y_t - y_{t-1} = (1 - L) y_t$ follows a stationary and invertible ARMA(p, q). Parameter d is equal to 1, and means, that ARMA(p, q) series are differenced once.

Examples of ARIMA(1,1,1) artificial processes



Examples of ARIMA(p,d,q) artificial processes



Estimation

Box and Jenkins (1970) were the first to approach the task of estimating an ARMA model in a systematic manner. There are 3 steps to their approach:

1. Identification
2. Estimation
3. Model diagnostic checking

The Information Criteria for Model Selection

The information criteria vary according to how stiff the penalty term is.

The three most popular criteria are Akaike's (1974) information criterion (AIC), Schwarz's (1978) Bayesian information criterion (SBIC), and the Hannan-Quinn criterion (HQIC):

$$\text{AIC} = \ln(\hat{\sigma}^2) + \frac{2k}{T},$$

$$\text{SBIC} = \ln(\hat{\sigma}^2) + \frac{k}{T} \ln(T)$$

$$\text{HQIC} = \ln(\hat{\sigma}^2) + \frac{2k}{T} \ln(\ln(T))$$

where $k = p + q + 1$, T = sample size.

Thus we choose the order of p and q according to **minimal Information Criteria**.